

Evidence-Weighted Validation of a Youth Athletic Development Philosophy

VERDICT (Bottom Line Up Front)

This is a well-founded, above-average philosophy: its central architecture — development over performance, dosing to biological maturity rather than chronological age, quality over fatigue, and treating enjoyment/confidence as load-bearing — sits squarely inside current consensus (NSCA 2016 LTAD position stand; Lloyd & Oliver's Youth Physical Development model; the 2014 BJSM International Consensus on youth resistance training). A coach running this would not be doing anything the evidence considers reckless, and several tenets (sprint/jump/lift as core, eccentric strength for hamstring protection, protecting speed quality) are genuinely well-supported.

But it has three real vulnerabilities the coach should confront:

1. **The "sprint mechanics are largely self-correcting / let force fix the form" tenet is the weakest link.** It overstates a genuine truth (strength transfers to speed) into a claim the science does not support. Strength transfer to sprinting is real but modest (mean sprint improvement 3.11%; Seitz et al. 2014), and elite sprint researchers (Haugen et al. 2018, 2019) treat technique as a distinct, trainable, coachable determinant, not a byproduct of force. This is a defensible stylistic lean toward external-cue/constraints-led coaching, but stated as a near-absolute it goes beyond the evidence.
2. **The "not a novice, so skill acquisition is largely handled" framing underweights motor-skill work.** The FMS literature shows a large share of children — even active ones — never reach fundamental movement skill proficiency, and models like Seefeldt's "proficiency barrier" and the YPD model keep FMS central through adolescence. An 8-year-old with a "training base" is not necessarily past this.
3. **The specific dosing numbers (40–60+ yard max-velocity sprints; heavy single-leg plyo emphasis) are asserted more precisely than the evidence licenses, and the near-total absence of a maturity-monitoring framework (PHV tracking, maturity offset, wellness/load monitoring) is a meaningful omission** given that the philosophy's own "ease dose through PHV" rule depends on knowing where PHV is.

None of these is disqualifying. But the philosophy is stated with more certainty than the underlying science in exactly the places where the field is still arguing, and the coach asked to have those places surfaced.

PART 1 — STEEL-MAN THEN STRESS-TEST EACH TENET

Tenet 1: "Sprint, Jump, Lift" as the unchanging three pillars for all youth at every age; structure constant, only dose changes.

Verdict: Aligns with consensus on content; partially conflicts on the "unchanging structure, only dose changes" absolutism.

Steel-man: sprinting, jumping (plyometrics) and resistance training map almost exactly onto the trainable qualities the YPD model and NSCA LTAD position stand prioritize across childhood and adolescence. The 2014 BJSM International Consensus (Lloyd, Faigenbaum, Stone, Oliver et al., *Br J Sports Med* 2014;48:498–505) explicitly endorses resistance training and high-velocity movement in youth. The idea that these three

qualities belong at every stage is consistent with the YPD model's core claim that "most, if not all, components of fitness are trainable throughout childhood" (Lloyd & Oliver 2012, *Strength Cond J* 34(3):61-72). (Substack)

Where it's contested: the claim that the *structure stays constant from youngest beginner to professional and only dose changes* runs against the emphasis of both YPD and the NSCA stand, which is that the *relative emphasis* of qualities should shift with maturation — FMS and motor-skill work weighted heavily pre-PHV, strength/power emphasized more around and after PHV. Lloyd & Oliver represent this with font-size changes across development, not a fixed template. The NSCA position stand explicitly warns that "any model should not be viewed as a stringent blueprint that can be superimposed on any participant." So "sprint, jump, lift at every age" is fine as a menu; "the structure never changes, only the dose" is a stronger claim than the maturation-based models make, and it specifically undersells the shifting-emphasis and skill-acquisition dimension. (National Strength and Con...)

Tenet 2: Strength-and-power-first, sprint mechanics largely self-correcting; minimal cueing ("let force fix the form").

Verdict: This is the tenet in most tension with the evidence. Partially aligns (strength does transfer) but the "self-correcting / let force fix form" framing conflicts with sprint-science consensus.

Steel-man: there is genuine support that strength underpins speed. Seitz et al. (2014, *Sports Medicine* 44(12):1693-1702; 15 studies, 510 subjects, 85 effect sizes) found "a very large significant correlation ($r = -0.77$; $p = 0.0001$) between squat strength ES and sprint ES," and Weyand's work established that top running speed is governed primarily by ground force, not faster leg turnover. For genuinely weak, immature athletes, raising force capacity plausibly does the most good, and the modern skill-acquisition/constraints-led school does favor minimal internal cueing and letting technique emerge from a well-designed task.

Where it conflicts: the same Seitz meta-analysis found the *average* sprint improvement from getting stronger was only 3.11% (sprint ES = -0.87) — real but modest, and it does not establish that force gains alone maximize speed. More directly, the elite sprint literature treats technique as a distinct, trainable, coachable determinant. Haugen et al. (2019, *Sports Medicine - Open* 5:44; DOI 10.1186/s40798-019-0221-0) state plainly that "key underlying determinants (e.g., power, technique, and sprint-specific endurance) are trainable" and devote a dedicated technical-training method section. Haugen et al. (2018, *Int J Sports Physiol Perform* 13(4):420-427; DOI 10.1123/ijsspp.2016-0812) found specific front-side kinematic variables — thigh angle at lift-off ($r = .64$) and maximal thigh extension ($r = .66$) — "largely correlated" with acceleration performance, and gave a coaching-relevant recommendation that "lower-leg back-lift should be minimised." No high-quality source states mechanics are "largely self-correcting once force/tendon stiffness improves." (springer)

The genuine nuance that partly rescues the coach: Haugen also cautions that "athletes cannot apply sprinting mechanics that they are not adequately predisposed to" — you can't cue a kid into a stride length his strength can't support, and heavy strength training "may induce negative short-term effects on sprint performance." So there is a real middle ground, and the debate between prescriptive/model-based cueing (Ralph Mann's front-side mechanics) and constraints-led/external-cue coaching (which favors self-organization) is genuinely unsettled. The coach's minimal-cueing preference is a *defensible stylistic choice* within that live debate. What is not defensible is the near-absolute framing that mechanics don't need coaching — that overstates one side of an unsettled argument as if it were settled. (ResearchGate)

Tenet 3: Dose scaled to biological maturity, ease dose through PHV.

Verdict: Strongly aligns with consensus — this is the philosophy's strongest tenet.

This is essentially the founding principle of the maturation-based models. The NSCA LTAD stand, the YPD model, and the BJSM consensus all reject chronological-age prescription in favor of biological maturity. Injury-surveillance data support easing load around PHV: growth-related conditions such as Osgood-Schlatter disease — documented at a 9.8% prevalence in adolescents aged 12–15 (11.4% in males, 8.3% in females) in the Brazilian population study by de Lucena et al. (*Am J Sports Med* 2011), and up to ~21% among sports-active adolescents — cluster around the circa-PHV window, with Rathleff and colleagues noting "approximately 10% of adolescents are affected by OSD, with a higher prevalence among those who are very active." Reviews in youth soccer (e.g., *J Sport Health Sci*, "Maturity-associated considerations... One size does not fit all") document elevated gradual-onset/overuse injury in the U12–U14 growth-spurt window. The one caveat is operational, not conceptual (see Part 2): the tenet is only as good as the coach's ability to actually locate PHV, and the philosophy provides no monitoring method.

Tenet 4: Real max-velocity sprinting (95–100%, 40–60+ yards) essential and non-substitutable; progress by volume/distance over months, not by chasing the stopwatch.

Verdict: The principle aligns; the specific distances (40–60+ yards) are not well-supported for youth max-velocity work and may be counterproductive.

Steel-man: the insistence on true high-intensity sprinting (not just drills or short bursts) is well-founded. Charlie Francis's "high-intensity meters" principle and Haugen's reviews agree that specific high-speed running is the primary specific stimulus for speed and cannot be fully replaced by strength or drills. Not chasing weekly PBs is sound — it protects mechanics and reduces the pressing that raises injury risk.

Where it's contested/risky: the 40–60+ yard prescription conflates *build-up distance* with *max-velocity exposure*. Haugen et al. (2019) note athletes sustain true maximal velocity for only ~10–30m even at elite level; young/untrained athletes are guided toward roughly a 20m build-up plus a 10m flying sprint, whereas elites might use a 40m build-up into a 30m fly. A flat-out 40–60+ yard youth sprint drifts into speed-endurance/deceleration territory, which carries higher fatigue and hamstring load. Because eccentric hamstring demand rises sharply above ~80% of max velocity, and because rapid ramps in high-speed volume are a documented hamstring-injury risk (return-to-run guidance recommends ~5% increments and avoiding large spikes), long flat-out sprints for youth are the opposite of the low-risk approach the philosophy otherwise espouses. The "progress by adding volume/distance" instruction also cuts against the more consensus practice of ending speed sessions on technical/velocity drop-off rather than accumulating distance. Note the ACWR "spike = injury" model is itself contested (Bowen, Fanchini, Lolli found weak predictive validity), so I present the hamstring-risk mechanism as biologically plausible and partly supported, not settled. (bluetoad) (PubMed Central)

Tenet 5: Plyometrics in all three planes, heavy single-leg emphasis; landing/deceleration trained inside the jump.

Verdict: Multi-planar and integrated-landing elements align well; the "heavy single-leg emphasis" is a defensible preference rather than an evidence-mandated one.

Steel-man: plyometric training is well-validated in youth. A maturity-stratified meta-analysis (Ramirez-Campillo/Moran et al., *Sports Medicine – Open* 2023;9:23; 744 participants, 11 studies) found plyometric jump training improves strength, sprint, jump and reactive strength with "a minimal exercise dosage of 4

weeks (8 intervention sessions), and 92 weekly jumps," and is "similarly effective in pre- and post-PHV youth in most outcomes, except for COD speed (in favour of pre-PHV)." Neuromuscular training that includes multi-planar plyometrics and landing mechanics reduces lower-extremity injury: Emery et al.'s dose-response meta-analysis of 16 trials (*Front Physiol* 2017;8:920) reported "an overall risk reduction of 42% with neuromuscular training (IRR = 0.58, 95%CI 0.47–0.72)." The single-leg/multi-planar logic ("sport is played on one leg") matches the mechanics of cutting and landing and is exactly the rationale of ACL-injury-prevention programs. Training the "stick"/deceleration inside the exercise is consistent with the specificity and skill-acquisition literature.

Where it's a preference, not a mandate: no meta-analysis establishes that a *single-leg-dominant* plyometric diet is superior to a balanced bilateral/unilateral program for youth. Most youth plyometric evidence used mixed or bilateral protocols. Single-leg landings also concentrate load — reasonable for late-stage/return-to-sport athletes, but the "heavy" emphasis for all youth is a coaching philosophy, not a settled finding. Dosing consensus is also softer than the tenet implies: youth basketball data (Frontiers 2024) suggest >150 jumps/week is needed to move sprint/COD, while other analyses find benefit at lower volumes; optimal ground-contact frequency estimates vary widely (e.g., <900 contacts favouring CMJ vs >1400 for SJ in one adolescent meta-analysis). (PubMed Central)

Tenet 6: Session order — sprint/jump first while fresh; never place top-speed sprint after a heavy plyo block; lifting closes.

Verdict: Aligns with the general principle (train speed fresh); the "never after plyos" absolute is oversimplified — plyometrics can potentiate as well as fatigue.

Steel-man: placing max-velocity and CNS-intensive work early, when fresh, is standard and sound. A fatigued athlete cannot express 95–100% effort, and quality speed work demands full recovery.

Where it's oversimplified: the relationship between plyometrics and subsequent sprinting is dose-dependent, not uniformly negative. The post-activation performance enhancement (PAPE) literature shows *low-volume, short* plyometric stimuli can *improve* subsequent sprint times (e.g., alternate-leg horizontal bounding improved 5m and 10m sprint by ~1.7% and ~1.1%; loaded jumps in warm-ups potentiate sprints), while *high-volume* jumping induces fatigue that degrades speed. So "sprint is never downstream of jumping" is true for a heavy plyo block but false as a blanket rule — a small, sharp plyometric primer before sprinting is a recognized potentiation strategy. The tenet is safe as a default but is stated more absolutely than the evidence supports.

Tenet 7: Eccentric/isometric-biased strength (Nordics, decel step-ups, iso holds) as a primary tool.

Verdict: Aligns for hamstring-injury prevention; "primary tool" is a defensible emphasis but the evidence base is more contested than commonly stated, especially in youth.

Steel-man: the Nordic hamstring exercise is among the most-studied injury-prevention tools. Van Dyk, Behan & Whiteley (2019, *Br J Sports Med* 53(21):1362–1370; 15 studies, 8,459 athletes, 525 injuries) found programs including the NHE roughly halved hamstring injuries, reporting "a reduction in the overall injury risk ratio of 0.49 (95% CI 0.32 to 0.74, p=0.0008)." Eccentric strength and decelerative capacity are directly relevant to the sprinting and cutting this philosophy centers on.

Where it's contested: (a) the effect has been challenged on methodological grounds — a reappraisal (Impellizzeri et al. 2021) found the preventive effect "inconclusive" once only true NHE-randomized RCTs were included, because many pooled studies bundled NHE with other exercises. (b) The optimal dose (low

vs high volume) is explicitly unresolved (Cholp & Zemková, *Front Physiol* 2025 systematic review). (c) Most NHE evidence is in adult/adolescent soccer, not young children; a youth football RCT (MDPI *Applied Sciences* 2024) found NHE training produced no significant sprint-performance change. So using eccentric/iso work is well-justified, but elevating it to the "primary" strength tool for youth — over foundational multi-joint strength and general resistance training that the NSCA/BJSM consensus put first — is a stylistic emphasis that runs slightly ahead of the youth-specific evidence.

Tenet 8: Conditioning is low priority because the sport supplies the stimulus.

Verdict: Partially aligns for many field/court sports, but contains a genuine physiological blind spot and is context-dependent.

Steel-man: for young athletes already playing multi-directional intermittent sports several times a week, dedicated aerobic conditioning is low-yield, and the sport does supply substantial stimulus. Prioritizing speed/strength quality over "junk" conditioning volume is reasonable and reduces overuse load.

Where it's contested: the assumption that the sport *adequately* conditions is not universally true and depends heavily on the sport, position, and playing time. There is also a live debate about whether preadolescents show a blunted aerobic trainability ("trigger hypothesis," Rowland; children's VO₂max "trainability deficit"), with evidence on both sides — some studies show meaningful preadolescent VO₂max gains (~8-15%; e.g., Obert et al., Baquet et al.), others minimal. The practical point: treating conditioning as categorically low priority risks under-preparing athletes in sports with high aerobic demand, and ignores that well-developed aerobic capacity aids recovery between the high-quality efforts this philosophy prizes. It's a reasonable default for many sports, but "the sport handles it" is an assumption to verify per athlete, not a law.

Tenet 9: Quality over fatigue — end sets when quality drops; short, fresh, precise sessions.

Verdict: Strongly aligns with consensus.

This matches the autoregulation/velocity-based-training logic (ending work on velocity loss), the sprint-science recommendation to end speed sessions on technical or velocity drop-off (Haugen 2019: "the session should be ended when drop-off in performance and/or technical deterioration is observed"), and the youth-training principle that neural/coordination adaptations are best trained fresh. It also aligns with the NSCA youth guidance on longer rest intervals (≥ 2 -3 min for multi-joint work) to preserve force/power quality in children. This is one of the philosophy's best-supported tenets and has essentially no credible opposition for developmental training.

Additional principles (task-specific warm-up ramps, rotating variations within a fixed structure, 9-10h sleep, pulling volume on game/practice days) are all consistent with best practice — RAMP-style warm-ups (Jeffreys), load-management logic, and the well-established role of sleep in youth recovery and injury risk. No conflicts here.

PART 2 — GAPS AND BLIND SPOTS

Motor-skill acquisition / "windows of trainability" and the "not a novice" framing.

Two things need separating. First, the philosophy is *right* to be skeptical of rigid "windows of trainability/opportunity" — that is exactly the critique Lloyd & Oliver (and Ford et al., *J Sports Sci* 2011)

leveled at the Balyi LTAD model, which was faulted because "there is a lack of empirical evidence upon which the model is based, particularly related to 'windows of opportunity.'" The YPD model was built to replace the strong-form windows claim with "most qualities are trainable throughout childhood." So if the philosophy rejects hard windows, it aligns with the modern view.

But the "already trained, so not a novice on skill" framing is a genuine blind spot. Being sport-experienced is not the same as being movement-competent. The FMS literature is blunt: a systematic review of FMS mastery (Br J Sports Med / PMC12517006) found many children reach adolescence without mastering most fundamental movement skills, and multiple international datasets (Germany, Australia, Finland, Belgium) show FMS proficiency is *declining* over time. Seefeldt's "proficiency barrier" (tested longitudinally, PMC9222670) posits that children below a competence threshold struggle to acquire more complex sport-specific skills — and the YPD model keeps FMS in the picture right through adolescence. The implication: an 8-year-old with several seasons of one sport may still have real gaps in general motor competence (throwing, catching, rotational and locomotor patterns outside his sport), and the philosophy's sprint/jump/lift chassis, while excellent for physical qualities, does not explicitly program broad motor-skill enrichment. That is precisely what the IOC youth consensus means by exposing even specialized youth to "the full menu of fundamental and foundational movement skills."

Overuse injury and early specialization.

The philosophy's multi-sport, development-first stance is well-aligned with the AOSSM (2016), AAP, and IOC (2015) consensus statements, which converge on: early single-sport specialization is not required for elite success and is associated with elevated overuse injury and burnout. The AOSSM think tank found "no evidence that young children will benefit from early sport specialization in the majority of sports... They are subject to overuse injury and burnout from concentrated activity." Useful concrete thresholds the philosophy could adopt explicitly: the commonly cited caution against training more hours per week than the child's age, keeping intense organized sport to <16 hours/week, and Jayanthi's specialization risk-classification work. The philosophy implies these but doesn't name volume thresholds — worth making explicit. (Note the field is refining "early" toward "premature" specialization — Bergeron, IOC — to center individual readiness over an age cutoff.)

Relative age effect (RAE) and maturity-status effects.

This is a real gap. The philosophy's "scale dose to maturity not age" handles the *training-prescription* half of maturation well, but says nothing about the *identification/selection* half. RAE (over-selection of relatively older children within an age band) and maturation bias (over-selection of early maturers) systematically distort who gets identified as "talented" and who gets more coaching and playing time — with early maturers favored and late maturers (who often have better long-term potential — the "underdog"/reversal effect documented by Cumming et al. 2018) filtered out. Bio-banding (Cumming, Lloyd, Oliver, Eisenmann, Malina 2017, *Strength Cond J* 39(2):34-47) is the mainstream mitigation, and the maturity-offset methods below make it actionable; a systematic review of interventions (SportRxiv) notes bio-banding and player-labelling can moderate but not fully remove the bias. A development-first philosophy that doesn't explicitly guard against maturity/RAE bias risks unknowingly over-rewarding the early-maturing kid and under-developing the late bloomer. "Scale dose to maturity" is necessary but, on its own, too simplistic to address selection bias.

Long-term athlete monitoring — the biggest structural omission.

The philosophy's own Tenet 3 ("ease dose through PHV") logically requires knowing where each athlete is

relative to PHV — yet no monitoring method is mentioned. Best practice, none of which appears in the philosophy:

- **Growth tracking:** serial standing and sitting height to detect the growth spurt directly (the most reliable practical signal).
- **Maturity offset / predicted age at PHV:** Mirwald et al. (2002) equations, with the caveat they lose accuracy for early/late maturers and away from PHV (mean absolute error roughly 0.5-1.8 years); the Moore et al. (2015) and Fransen et al. (2018) equations are now generally preferred, and Khamis-Roche (percentage of predicted adult height) is favored for maturity status. These are population tools, not precision instruments.
- **Wellness questionnaires and training-load monitoring** (sRPE, ACWR — with the caveat that ACWR's predictive validity is contested), plus monitoring for growth-related conditions (Osgood-Schlatter, Sever's) around PHV.

The absence of any monitoring framework is a meaningful gap because it turns the philosophy's best principle (maturity-based dosing) into something the coach can only eyeball. Adding simple growth tracking would materially strengthen the whole approach.

State of evidence on youth plyometric/sprint dosing (specific numbers).

Consensus exists that plyometrics and sprinting benefit youth; consensus does *not* exist on the specific volumes/distances the philosophy asserts. Plyometric dose-response estimates vary widely (minimal effective dose ~92 jumps/week in the Moran/Ramirez-Campillo youth meta-analysis; >150 jumps/week to move sprint/COD in youth basketball; divergent optimal ground-contact estimates). For max-velocity sprinting, as noted, the 40-60+ yard flat-out prescription is not specifically supported for youth and likely exceeds the true max-V zone; ~2 high-quality speed sessions/week with ≥48h between, ended on drop-off, is the better-supported model (Haugen: intensive sprint sessions "require at least 48 h of recovery... sprinting on consecutive days rarely occurs among leading practitioners"). So the philosophy's numbers are within a defensible range for plyometric emphasis but overstated for sprint distance, and in both cases are presented with more precision than the literature justifies.

Genuine risk flags (as opposed to stylistic differences).

1. **Long flat-out 40-60+ yard youth sprints:** the clearest injury-relevant concern — high eccentric hamstring load, drifts beyond the max-velocity zone, and rapid high-speed-volume ramps are a plausible hamstring-injury driver. Recommend flying-sprint structure (short build-up + ~10-30m fly) and progressing ~5% increments.
2. **No maturity-monitoring method** despite a maturity-dosing rule: the coach could miss the PHV window they intend to protect.
3. **Heavy single-leg plyo emphasis** without individual load monitoring concentrates joint load; fine with progression and landing quality, riskier if volume climbs unmonitored.

Everything else (minimal sprint cueing, session ordering, eccentric-primary emphasis, low-priority conditioning) is a defensible stylistic/philosophical stance within live debates — not an evidence violation.

PART 3 — THE BOMPA BOOK

(a) What *Training and Conditioning Young Athletes*, 2nd ed. (Bompa & Sarandan, Human Kinetics, 2023) actually is.

Verifiable facts (Human Kinetics catalog, VitalSource, Library of Congress data via the book's front matter): the book is by Tudor O. Bompa and Sorin O. Sarandan, published by Human Kinetics, ©2023, 304 pages, ISBN 9781718216143. Per the Library of Congress record printed in the book, it is "a revised edition of *Conditioning Young Athletes*, published in 2015 by Human Kinetics" (the 2015 first edition was co-authored by Bompa and Michael Carrera). It advertises 200+ resistance and conditioning exercises organized for "six stages of youth training," accounting for developmental stage, motor function, and sex-specific considerations, plus nutrition guidance and an analysis of energy systems for 13 sports. (Human Kinetics + 4)

Verifiable table of contents (Human Kinetics product page):

- **Part I. Training Considerations** — Ch. 1 Effects of Long-Term Training on Young Athletes; Ch. 2 Injuries and Injury Prevention; Ch. 3 Guidelines for Long-Term Training; Ch. 4 Stages of Long-Term Athletic Development.
- **Part II. Nutrition and Energy System Training** — Ch. 5 Nutrition Considerations; Ch. 6 Energy System Training for Sport; Ch. 7 Model Training for Team Sports.
- **Part III. Training Methods** — Ch. 8 Science-Based Strength Training Methodology; Ch. 9 Training Methods to Develop Strength and Power; Ch. 10 Training Methods to Develop Speed; Ch. 11 Training Methods to Develop Agility and Flexibility; Ch. 12 Training Methods to Develop Specific Endurance.
- **Part IV. Selected Exercises** — Exercise Finder.

What is *inferred* from Bompa's broader body of work (not verified line-by-line in this edition, and flagged as such): the book's LTAD framework rests on Bompa's long-standing principles — a base of **multilateral (multi-skilled) development** before specialization (his classic training pyramid; a Scribd copy of the front matter reproduces his statement that "children and youth who develop a variety of skills and motor abilities are more likely to adapt to demanding training loads without experiencing stresses associated with early specialization"), and periodization/annual-plan structure from *Periodization: Theory and Methodology of Training*. Published Human Kinetics excerpts from this edition ("Long-Term Periodization of Training," "Modeling the Microcycle: Three Suggested Training Models") confirm periodization and microcycle planning are central. The specific contents of the "six stages" and the exact recommendations within each chapter should be verified directly by the user via the Human Kinetics page, Google Books preview, or a library copy (WorldCat) — I have not seen a verified full-text chapter breakdown beyond the TOC above, and will not fabricate one.

(b) How Bompa's framework relates to the philosophy.

Strong agreement:

- **Multilateral development before specialization** ↔ the philosophy's broad-foundation, development-over-performance, multi-sport stance. This is the deepest point of alignment.
- **Injury prevention and long-term orientation** ↔ the philosophy's durability-over-a-decade framing.
- **Sequenced development of strength, power, speed, agility, endurance** ↔ sprint/jump/lift plus the recognition that qualities are built over years.

Tension/conflict:

- **Periodization structure vs. "fixed structure, rotate variations."** Bompa's hallmark is the periodized annual plan with distinct phases; the philosophy keeps a *constant* session structure and rotates exercise variations. These aren't incompatible, but they emphasize different things — Bompa plans macro-cycle variation; the philosophy holds structure constant and varies micro-content.
- **Stage/age-based staging vs. maturity-based dosing.** Bompa's LTAD is organized into stages (here, "six stages"), which historically lean on chronological-age anchoring; the philosophy is explicitly maturity-driven. This is the same fault line the field draws between older phase models and the newer YPD model (below).
- **Conditioning.** Bompa devotes whole chapters to energy-system and specific-endurance training for youth; the philosophy deprioritizes conditioning. Direct point of divergence.

(c) How Bompa is regarded now, and critiques of phase/age-based youth periodization.

Bompa is genuinely a foundational figure — the "father of periodization," NSCA's 2014 Alvin Roy Career Achievement awardee, and his texts are field-standard. His multilateral-development message is fully consistent with modern consensus. (Human Kinetics)

However, the specific *critique relevant to youth* is real and mainstream: rigid, phase-based, chronological-age-anchored periodization models are criticized for not accommodating the highly individual, non-linear timing of growth and maturation. This is precisely the critique Ford, Lloyd, Oliver and colleagues (*J Sports Sci* 2010/2011) leveled at the Balyi LTAD model — that age-based classification "seems to be inherently flawed due to variation in growth and maturational rates between individuals" and that the "windows of opportunity" concept lacked empirical support. Lloyd & Oliver's YPD model (2012) was the explicit response: replace age-based staging and hard windows with maturity-based, concurrent development of all qualities. The NSCA LTAD position stand (2016) codified this, warning against treating any model as "a stringent blueprint." Bompa's underlying periodization theory has also drawn broader scientific criticism (e.g., from Kiely) for resting on dated general-adaptation-syndrome assumptions and limited RCT support.

Net: the philosophy under review is, on the age-vs-maturity question, *closer to the modern (YPD/NSCA) position than to Bompa's classic phase model* — its maturity-based dosing is more current than Bompa's stage-based staging. Where the philosophy could learn from Bompa is the reverse: his stronger commitment to genuine multilateral/motor-skill breadth and to structured periodized variation, both of which the philosophy handles more thinly.

RECOMMENDATIONS (staged, concrete)

Keep as-is (well-founded): maturity-based dosing, quality-over-fatigue, speed-first session order (as a default), multi-planar plyometrics with integrated landings, eccentric hamstring work, sleep and game-day load reduction, task-specific warm-ups.

Fix now (evidence says so):

1. **Replace flat-out 40–60+ yard youth sprints with flying-sprint structure** — short build-up (~20m) into a ~10–30m maximal segment, ≤ 2 quality speed sessions/week, ≥ 48 h apart, ended on velocity/technical drop-off. Progress high-speed volume in ~5% increments. Benchmark to relax this: per-athlete data show tolerance and no hamstring flare.

2. **Add a minimal maturity-monitoring layer:** quarterly standing + sitting height (growth-velocity tracking) and a maturity-offset/percentage-of-adult-height estimate (Khamis-Roche, or Moore/Fransen), plus a short weekly wellness check. This operationalizes the philosophy's own PHV rule.
3. **Soften the two absolutes:** "let force fix the form" → "prioritize force, but coach mechanics when a technical feature (e.g., front-side positioning) is limiting"; "never sprint after plyos" → "never after a heavy plyo block; a small sharp plyometric primer can potentiate speed."

Add (fills gaps): 4. **Explicit general motor-skill/FMS enrichment**, even for the "trained" 8-year-old — throwing, catching, rotational and multidirectional patterns outside the primary sport. Don't assume sport experience equals movement competence. 5. **Guard against RAE/maturation bias** in any selection/attention decisions; consider bio-banding logic for training and games. 6. **Verify the conditioning assumption per sport/position** rather than treating it as categorically low priority; retain some aerobic base for recovery between high-quality efforts. 7. **Name specialization/volume thresholds** (weekly hours $\leq$ age; $<16\text{h}$ /week intense; $\geq 1\text{--}2$ rest days) to make the multi-sport stance concrete.

Benchmarks that would change the recommendations: emergence of youth-specific RCTs establishing single-leg-dominant plyometric superiority (would upgrade Tenet 5 from preference to mandate); youth-specific evidence that long max-V sprints are safely tolerated (would relax rec. 1); resolution of the NHE dose and youth-transfer debate (would firm up Tenet 7).

CAVEATS

- Much of the strongest youth-specific evidence is in soccer and track populations; transfer to other sports is reasonable but not guaranteed.
- Several dose-response questions (plyometric volume, NHE dose, ACWR predictive validity, preadolescent aerobic trainability) are genuinely unresolved — I have flagged these as debates, not settled points.
- The van Dyk NHE effect and the neuromuscular-training injury-reduction figures come largely from adolescent/adult team-sport samples; extrapolation to young children is an inference, not a demonstrated result.
- The Bompa 2nd-edition chapter-level contents beyond the verified TOC could not be confirmed without the book itself; the user should verify specific "six-stage" recommendations via Human Kinetics, Google Books, or a library copy rather than rely on inference from Bompa's broader work.
- Practitioner sources (SimpliFaster, coaching blogs) were used only for convention/context and are flagged as non-peer-reviewed; all load-bearing claims rest on position stands, meta-analyses, and peer-reviewed reviews.